

University
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SUBSPACE IDENTIFICATION METHODS FOR EXTRACTING BEHAVIORALLY RELEVANT NEURAL DYNAMICS

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Introduction

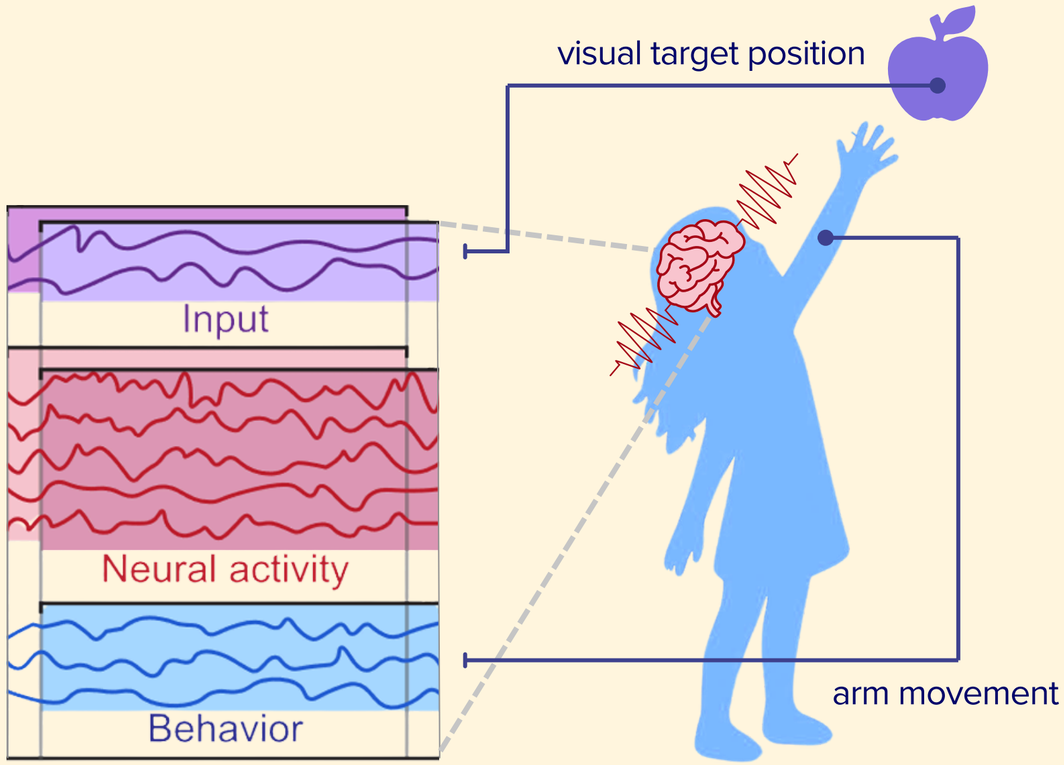
Understanding neural signals is important:

➜ *brain theories*

➜ practical applications: *brain-machine interfaces*

However, **brain activity** is the result of **different contributions**:

- ***input-driven*** activity
- ***behavioral*** neural activity
- ***other*** neural activity



Example: a human reaching for an apple

Modeling Neural Data

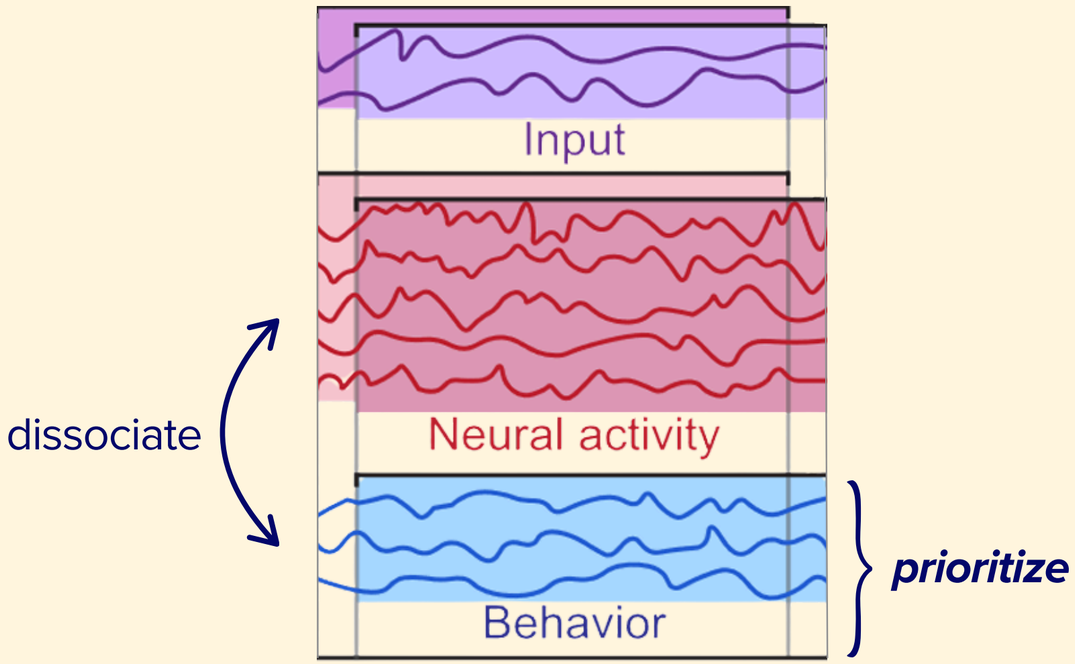
1

Prioritize Behavior *(Objective 1)*

Dissociate and ***prioritize behaviorally relevant*** dynamics from other dynamics

Why?

- Most neural variance is behaviorally irrelevant
- No prioritization → model captures dominant (not behavior-predictive) dynamics



Modeling Neural Data

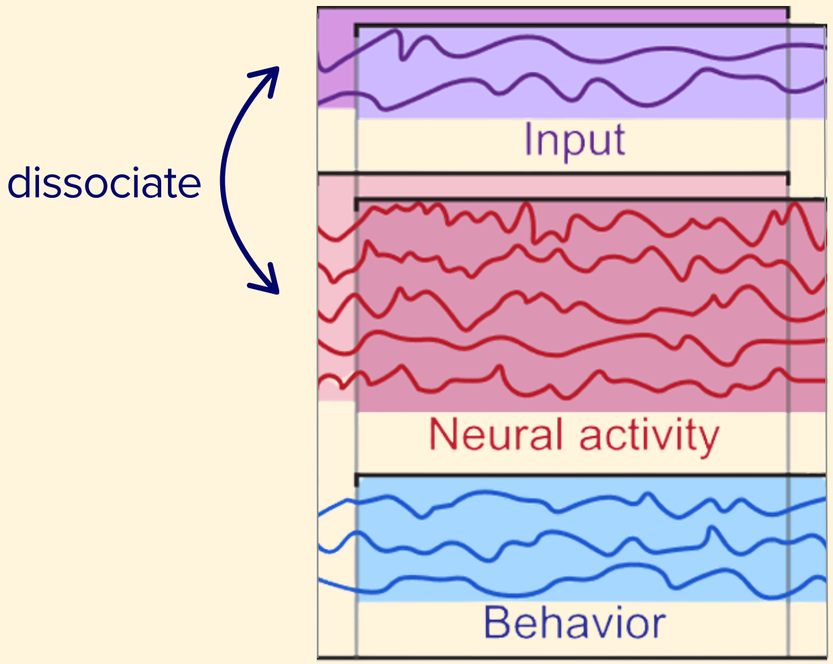
2

Dissociate Input *(Objective 2)*

Dissociate intrinsic neural dynamics ***from input-driven*** dynamics

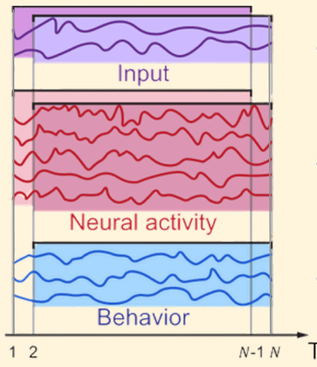
Why?

- Inputs carry their own temporal structure
- No dissociation ➜ input dynamics are mistaken for intrinsic ones



General Framework

Measurements



- **u_k input**
- **y_k neural activity**
- **z_k behavior**

State-space model

The measurements evolution over time is modeled via a **dynamical system**

Non-preferential methods (NDM, INDM)

Preferential methods (PSID, IPSID)

$$\begin{cases} x_{k+1} = Ax_k + Bu_k + w_k, \\ y_k = C_y x_k + D_y u_k + v_k, \\ z_k = C_z x_k + D_z u_k + \epsilon_k \end{cases}$$

x_k latent state → low dimensional vector encoding neural dynamics

Latent state extraction
+
System matrices estimation

Predict behavior

Given new neural recordings, **behavior can be predicted** without directly observing it